(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-112 ***Subject: Applied Mathematics-II***

Time: 3 Hours ***Maximum Marks: 60***

Note: Attempt any five questions including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt all of the following:- **(3x4=12)**

(a) Resolve $e^{\sin(x+iy)}$ into real and imaginary parts

(b) Find the image of infinite strip $\frac{1}{4} \le y \le \frac{1}{2}$. under the transformation $w = \frac{1}{z}$.

(c) Find the Laplace transform of $\frac{Cosat - Cosbt}{t}$.

(d) Using the method of Separation of variable , solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$

Where $(x, 0) = 6e^{-3x}$.

UNIT-I

Q2 (a) Find the value of C_1 and C_2 such that the function **(6)**

$f(z) = x^2 + C_1y^2 - 2xy + i(C_2x^2 - y^2 + 2xy)$ is analytics. Also find $f'(z)$.

(b) Show that the function $u = \frac{1}{2}\log(x^2 + y^2)$ is harmonic . Find its harmonic Conjugate. **(6)**

Q3 (a) State Cauchy Integral formula and hence evaluate **(6)**

$\int_C \frac{3z^2 + z}{z^2 - 1} dz$, where C is the circle $|z - 1| = 1$.

(b) Evaluate the Line Integral $\int_C z^2 dz$, Where C is the boundary of a triangle with vertices $0, 0 + i, -1 + i$, Clockwise . **(6)**

UNIT-II

Q4 (a) Evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$, Where C is $|z| = 3$ by using Cauchy residue theorem. **(6)**

(b) Evaluate Laurents series which represents the function **(6)**

$f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$ When (i) $|z| < 2$, (ii) $2 < |z| < 3$.

Q5 (a) Apply the calculus of residues to evaluate **(6)**

$\int_{-\infty}^{\infty} \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx$, $a > b > 0$.

(b) Let $f(z)$ be a bilinear transformation such that $f(\infty) = 1$, $f(i) = i$, *and* $f(-i) = -i$

Find the image of unit dick (c) $\{z \epsilon c; |z| < 1\}$ under (z) . **(6)**

P.T.O.

P-1/2
BS-112

(Please write your Enrolment No. immediately) Enrollment No .....

Mid Term Examination

Second Semester, April ,2024

Paper Code: BS 112 Subject: Applied Mathematics -II

Time $1\frac{1}{2}$ Hrs. Max. Marks:30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining .

Q		Max. Marks	CO(s)
1(a)	Convert the complex number $\frac{-16}{1+i\sqrt{3}}$ into Polar form .	2.5	1
(b)	Use Cauchy Integral Formula to evaluate $\int_c \frac{\text{Sin}\,\pi z^2 + \text{Cos}\,\pi z^2}{(z-1)(z-2)} dz$,where C is $\|z\|=3$	2.5	1
(c)	Find Invariant Point in $w = \frac{3z-4}{z-1}$	2.5	2
(d)	Determine the residue of function at each Pole. If, $f(z) = \frac{z^2}{(z-1)(z-2)^2}$.	2.5	2
2(a)	Find the Analytic function $f(z) = u + iv$, $Given\ that\ v = e^x(\text{xsin}\,y + y\cos y)$	5	1
(b)	Evaluate the Integral I = $\int_C \text{Re}\,(z^2)dz$ .From Z=0 to 2+4i along the Parabola $y = x^2$	5	1
3(a)	Show that $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{(a^2-b^2)^{\frac{1}{2}}}$, a> b>0 . Using Contour Integration.	5	2
(b)	Find the Laurent's expansion of $f(z) = \frac{1}{z(z^2-3z+2)}$ For the region $1 \leq \|z\| < 2$.	5	2
4(a)	Find the image of circle $x^2 + y^2 = 4y$ $Under\ the\ transformation\ w = \frac{1}{z}$.	5	2
(b)	Discuss the analyticity of the function $f(z) = z\bar{z}$ at origin.	5	1